

MIHAN

AI Freelancer Credit Scoring Engine

Full Compliance & Feasibility Assessment

Alinma Bank | AMAD Hackathon 2026 | July 16-18, Riyadh

6

Skills Run

34

Compliance Items

13

Legal Risks Scored

24

UX Findings

Overall Assessment

Mihan is an AI-powered B2B SaaS credit scoring engine enabling Alinma Bank to lend to Saudi freelancers — 1.6 million Freelance Work Document holders currently excluded from all commercial bank lending. The hackathon demo is fully buildable in 72 hours with zero regulatory blockers. Post-hackathon, the pilot path is clear but requires Alinma as an active partner, particularly for SAMA submissions.

VERDICT: Strong position to win. Highest risk is demo execution, not the idea.

THREE THINGS THAT COULD KILL MIHAN BEFORE IT STARTS

1. Data processed outside Saudi Arabia — criminal liability, 1 year imprisonment + SAR 1M fine. Architecture must enforce me-south-1.
2. No DPIA + DPO before pilot — criminal liability, 2 years + SAR 3M fine. Engage Saudi consultancy immediately post-hackathon.
3. SAMA income methodology unapproved — pilot blocker. Submit Worst-Month x 80% via Alinma the day after the hackathon ends.

1. SAUDI REGULATORY COMPLIANCE CHECK

The hackathon demo has no regulatory blockers — it uses synthetic data only. The pilot requires pre-work on SAMA income methodology and PDPL obligations before any real loan is issued.

Regulation	Demo	Pilot
SAMA Rule 300-1-1-4 (freelancer accounts)	N/A	Alinma already compliant
SAMA Consumer Financing (33.33% cap)	N/A	BLOCKER — submit Worst-Month x80% to SAMA via Alinma
SAMA Responsible Lending (SIMAH concurrent)	Mock OK	Must run live SIMAH in parallel with scoring
SAMA AI: disclosure, explainability, human review	Build modal	Same requirements carry forward to production
PDPL Data Residency	N/A (local)	CRIMINAL — AWS me-south-1 only, written confirmation required
PDPL DPIA + DPO	N/A	CRIMINAL — must be in place before any real data processed
PDPL Cross-product consent	Wire in UI	Must be live with immutable timestamped storage
PDPL Automated decision (privacy notice)	Not required	Required before any freelancer application
Credit Information Law (Proof of Income)	Demo fine	Frame as income certificate, not credit report
SAMA Open Banking Framework (Phase 2)	Not needed	Requires sandbox application — 9-12 month review
SDAIA AI Ethics Framework	Label AI outputs	Full audit trail + fairness audit before scale
ZATCA VAT	Not applicable	15% VAT on B2B invoices to Alinma before invoicing
Kafalah	Do not mention	Freelancers don't qualify — never reference in pitch or docs
Sharia / Tawarruq	One pitch slide	Alinma existing structures apply — no new Sharia approval

3 Items Confirmed Legally Sound

[OK] B2B fee structure (SAR 30-50 per assessment) — Alinma absorbs, never passed to borrower. 1% cap on SAR 60K loan = SAR 600 max.

[OK] Contractual standing deduction with prior written consent at signing — fully SAMA-compliant.

[OK] Phase 1 under Alinma's existing banking license — no new SAMA license needed for demo or pilot.

2. LEGAL RISK ASSESSMENT

13 legal risks scored using a Severity x Likelihood matrix. 4 Critical (RED), 3 High (ORANGE), 5 Medium (YELLOW), 1 Low (GREEN). The 4 critical risks are all pre-pilot, not pre-demo.

#	Risk	Sev	Like	Score	Level	RAG
1	PDPL data residency — processing outside KSA	5	4	20	Critical	● RED
2	PDPL automated decisions — DPIA + DPO missing	5	4	20	Critical	● RED
3	SAMA income cap — no approval for self-employed methodology	5	5	25	Critical	● RED
4	Cross-product consent violation — SDAIA enforcement	4	4	16	Critical	● RED
5	IP ownership of scoring model weights post-pilot	4	3	12	High	● AMBER
6	AI scoring liability — model approves loan that defaults	4	3	12	High	● AMBER
7	Force majeure — Mihan SaaS down, Alinma loan book affected	3	3	9	Medium	● AMBER
8	Digital Proof of Income grey area (certificate vs credit report)	3	3	9	Medium	● AMBER
9	Sharia non-compliance of loan structure	3	2	6	Medium	● AMBER
10	Consumer protection law applicability to freelancer borrowers	2	3	6	Medium	● AMBER
11	AML interaction with AI scoring layer	2	2	4	Low	● GREEN
12	Competition law — data moat seen as anti-competitive by SAMA	2	2	4	Low	● GREEN
13	Kafalah misrepresentation risk	1	2	2	Low	● GREEN

Mitigations for 4 Critical Risks

Risk 1 — Data Residency

Architecture must enforce Saudi region at infrastructure level. Claude API proxy sends only derived factor scores (0-100 numbers), never PII. Legally classified as aggregated statistics — document in DPIA.

Risk 2 — DPIA + DPO

Engage outsourced Saudi data protection consultancy (KPMG Saudi, Deloitte Saudi) immediately post-hackathon. Budget SAR 50K-150K. Cannot be deferred to pilot launch.

Risk 3 — SAMA Income Methodology

Only path is Alinma submitting Worst-Month x80% formula to SAMA. Start the conversation at the hackathon. No live loans can be issued without this approval — it is the single longest-lead item.

Risk 4 — Cross-Product Consent

Build explicit checkbox with immutable timestamped storage (append-only log, no modification allowed) before any real user touches the product. SDAIA issued 48 enforcement decisions in 2025 — actively policed.

3. OPERATIONAL RISK ASSESSMENT

11 operational risks assessed across hackathon execution and post-hackathon pilot phases.

Hackathon Risks (72-Hour Window)

Risk	Likelihood	Impact	Level	Mitigation
Demo crash during live pitch	Medium	Critical	HIGH	Pre-record 90-sec video; pre-cache all API responses; frozen HTML fallback screenshots
Synthetic data looks fake to judges	High	High	HIGH	Arabic names, realistic SAR amounts, real Saudi professions (App Developer, Graphic Designer)
Arabic UX errors in front of Alinma judges	Medium	High	HIGH	Native speaker review mandatory by July 14 — not optional
Live SIMAH API failure during demo	High	High	HIGH	Mock completely. Pre-rendered thin-file response is the correct product story
Team member unavailable in 72-hour window	Low	High	MEDIUM	Document all components; each person owns end-to-end delivery of one module
Scoring produces counterintuitive results on synthetic data	Medium	Medium	MEDIUM	Test all 3 profiles manually before demo; tune weights if needed
Judges challenge '4 tracks' scope claim	High	Medium	MEDIUM	Lead with Fintech track; others secondary; one-line rationale ready for each
Cloud latency during demo	Low	High	MEDIUM	Run scoring engine locally — zero cloud dependency for demo day
Pitch delivery overruns time	High	Medium	MEDIUM	Timed dress rehearsal at hour 60; 4-minute hard cap

Post-Hackathon Pilot Risks

Risk	Level	Mitigation
500-person onboarding friction	MEDIUM	Alinma branch staff pre-briefed; Arabic onboarding video prepared in advance
Standing deduction failures	MEDIUM	Contractual consent at signing; Alinma handles deduction mechanics
Early defaults in pilot cohort	MEDIUM	Conservative Green tier threshold (75+); Building tier gets roadmap, not a loan
SAMA asks for methodology revision	HIGH	Alinma relationship is the buffer; build full audit trail from day 1 of pilot

4. TECHNICAL ARCHITECTURE — ADR-001

Hackathon Demo Stack (Local, 72 Hours)

LOCAL LAPTOP

Next.js 14 (Arabic RTL, mobile-first)

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FastAPI (Python) — Scoring Engine

| - expense_discipline() x 0.35

| - income_stability() x 0.30

| - client_diversity() x 0.20

+ - savings_behavior() x 0.15

|

SQLite — synthetic profiles + append-only audit log

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Claude API Proxy (pre-cached) — factor scores only, zero PII

Mock SIMAH — pre-rendered JSON response

Pilot Stack (AWS me-south-1 / me-central-1)

AWS SAUDI REGION (KSA ONLY)

CloudFront -> Next.js (Amplify or ECS Fargate)

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API Gateway -> Lambda / ECS FastAPI

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RDS PostgreSQL (me-south-1, encrypted at rest)

S3 (me-south-1) — PDF bank statement storage

AWS Textract (me-south-1) — OCR, in-region only

|

Claude API Proxy — derived factor scores ONLY (no PII exits KSA)

Alinma Internal API — transaction pull (business account)

SIMAH Live API — concurrent check (mandatory, not replaced)

Nafath API — digital identity verification

Audit Log: append-only DynamoDB table

SAMA Inspector: read-only IAM role + dashboard

PDPL-Safe Claude API Pattern

The Claude API call receives ONLY derived factor scores (integers 0-100). No name, no IBAN, no transaction amounts, no dates, no employer names. Legally classified as aggregated statistics under PDPL — not personal data. Document this in the DPIA explicitly.

Feasibility Q&A

Question	Answer
Minimum viable hackathon stack?	Next.js + FastAPI + SQLite + pre-cached Claude. No cloud needed.
Additional pilot components?	RDS, S3, Textract, live SIMAH/Nafath APIs, consent storage, audit log
Biggest technical risk?	PDF bank statement parsing — solved by pdfplumber (demo) / AWS Textract in me-south-1 (pilot)
Claude API PDPL compliance?	Proxy layer — only factor scores leave Saudi cloud; legally aggregated statistics
PDF parsing in KSA?	AWS Textract available in me-south-1. Never parse client-side. Data stays in KSA.

5. ACCESSIBILITY & UX AUDIT — WCAG 2.1 AA

24 findings across 6 screens. 7 Critical (block WCAG compliance), 11 Major, 6 Minor. SAMA AI requirements also audited. All critical fixes required before demo day.

Critical Fixes (7 items — must be done before demo)

Screen	Issue	Fix
Decline Screen	Human review button contrast 2.9:1 — FAILS 4.5:1 minimum (WCAG 1.4.3)	Change to #0052A3 white text — 6.8:1 ratio. Alinma brand blue.
AI Disclosure Banner	Display-only banner — no acknowledgement (SAMA requirement)	Convert to modal with role='alertdialog' and confirm button
Score Card	Tier communicated by colour only — WCAG 1.4.1 failure	Add tier name text to aria-label on score gauge
Decline Screen	Claude explanation not labeled as AI-generated (SAMA + SDAIA)	Add '[AI-Generated Explanation]' label and AI badge
Consent Screen	Checkbox touch target 16px — minimum is 44x44px (WCAG 2.5.5)	Full-width <label> wrapping entire checkbox text
Banker Dashboard	Table missing dir='rtl' — column order wrong in Arabic	Add dir='rtl' + reverse column sequence in HTML
All Arabic screens	Numbers render LTR inside RTL text — BiDi rendering bug	Wrap all numbers in SAR 8,500

SAMA AI Compliance — Screen-by-Screen

SAMA Requirement	Current State	Status
Inform customer AI is used BEFORE assessment	Banner exists but no acknowledgement	FAIL — must convert to modal
Plain-language decline reasons (top 2-3 factors)	Factor cards present with scores	PASS
'Request Human Review' available for declines	Button present but below fold + contrast fails	FAIL — move to top, fix colour
AI-generated content labeled as AI	Not labeled anywhere	FAIL — add label to Claude paragraph
Full audit log accessible to SAMA inspectors	Not built yet (demo)	BUILD for pilot; mention in pitch

Code Fixes — Copy-Paste Ready

```
/* Human review button — was 2.9:1, now 6.8:1 */
.human-review-btn { background: #0052A3; color: #FFFFFF; }

/* Score gauge aria-label */
<div role="img" aria-label="Mihan Score: 79/100 - Green Tier">

/* Numbers in Arabic UI */
<p>Monthly income: <span dir="ltr">SAR 8,500</span></p>

/* AI disclosure modal */
<div role="alertdialog" aria-labelledby="ai-title" aria-modal="true">
  <h2 id="ai-title">Important Notice — يُرجى إكمال الإجابة</h2>
  <button onclick="acknowledgeAndProceed()">I understand, proceed</button>
</div>
```

6. COMPLIANCE TRACKING REGISTER — 34 ITEMS

RAG status: RED = blocking (cannot proceed), AMBER = at risk (must be done), GREEN = done. All items are currently RED or AMBER — none complete at time of assessment (2026-05-19).

PHASE 1 — Pre-Hackathon (All items due by July 15, 2026)

#	Item	Framework	RAG	Priority
1	AI disclosure acknowledgement modal (banner -> modal with confirm)	SAMA AI	● RED	BLOCKING
2	Fix human review button contrast (#FF8C00 -> #0052A3)	WCAG 1.4.3	● RED	BLOCKING
3	Add dir=rtl + column reversal to banker dashboard	WCAG RTL	● RED	BLOCKING
4	aria-label on score gauge including tier name text	WCAG 1.4.1	● AMBER	MAJOR
5	Full-width clickable label for consent checkbox	WCAG 2.5.5	● AMBER	MAJOR
6	Label Claude explanation as AI-generated	SAMA + SDAIA	● RED	BLOCKING
7	Move human review button to TOP of decline screen	SAMA AI / UX	● AMBER	MAJOR
8	Wrap all numbers in in Arabic UI	WCAG RTL	● AMBER	MAJOR
9	Add consent withdrawal to account settings	PDPL	● AMBER	MAJOR
10	Synthetic data: Arabic names, SAR, Saudi professions	Operational	● RED	BLOCKING
11	Mock SIMAH — never attempt live API	Operational	● RED	BLOCKING
12	Pre-cache Claude API responses for all 3 profiles	Operational	● RED	BLOCKING
13	Dress rehearsal with timed pitch at hour 60	Operational	● AMBER	CRITICAL
14	Arabic copy reviewed by native speaker (finance context)	Operational	● RED	BLOCKING
15	Sharia compliance framing added to pitch deck	Islamic Finance	● AMBER	IMPORTANT
16	Demo video pre-recorded as 90-sec fallback	Operational	● AMBER	INSURANCE

PHASE 2 — Pre-Pilot (Before Any Real User Data)

#	Item	Framework	RAG	Note
17	Confirm AWS me-south-1 or Azure Saudi North — written confirmation	PDPL Residency	● RED	CRIMINAL
18	DPIA completed and filed with SDAIA	PDPL Automated	● RED	CRIMINAL
19	DPO appointed (outsourced Saudi consultancy acceptable)	PDPL	● RED	CRIMINAL
20	Submit Worst-Month x80% income methodology to SAMA via Alinma	SAMA Consumer	● RED	PILOT BLOCKER
21	Explicit cross-product consent checkbox built into loan flow	PDPL Purpose	● RED	ENFORCED
22	Vendor agreement with Alinma signed (IP, liability, SLA, AML)	Contract	● RED	UNLOCKS PILOT
23	ZATCA VAT registration for B2B SaaS fees	ZATCA VAT	● AMBER	PRE-REVENUE
24	E&O insurance obtained for SaaS product	Liability	● AMBER	PRE-PILOT
25	Penetration testing of scoring engine and data pipeline	Security / PDPL	● AMBER	PRE-PILOT
26	Privacy notice updated for automated decision-making	PDPL	● AMBER	PRE-PILOT
27	Consent stored with immutable timestamp + user ID	PDPL Consent	● AMBER	SDAIA AUDIT
28	SAMA inspector read-only access to audit logs	SAMA AI	● AMBER	ALINMA MANAGES

29	Alinma Sharia Supervisory Board formal endorsement	Islamic Finance	● AMBER	ALINMA INTERNAL
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PHASE 3 — Post-Pilot / Phase 2 Scale

#	Item	Framework	RAG	Timeline
30	SAMA Regulatory Sandbox application (Open Banking credit scoring)	SAMA Open Banking	● AMBER	Q4
31	Saudi legal counsel retained (Al Tamimi / AlGhazzawi)	Regulatory	● AMBER	PRE
32	Digital Proof of Income legal framing confirmed by counsel	Credit Info Law	● AMBER	PRE
33	Consent withdrawal mechanism live in production	PDPL	● AMBER	PRE
34	Fairness/bias audit across 120+ professions and demographics	SAMA AI + SDAIA	● AMBER	POST

Master RAG Dashboard

Phase	Total	RED Blocking	AMBER At Risk	GREEN Done
Pre-Hackathon (by July 15)	16	7	9	0
Pre-Pilot (before real data)	13	6	7	0
Post-Pilot / Phase 2	5	0	5	0

7. DEMO BUILD SCOPE — WHAT TO BUILD IN 72 HOURS

Exact scope for the hackathon demo. Build these 5 modules and nothing more. Estimated ~42 hours of coding; 30 hours buffer for Arabic polish, testing, and pitch preparation.

Module 1 — Scoring Engine (Backend, ~10 hours)

FastAPI in Python. 4 factor functions with exact weights (35/30/20/15). 3 synthetic freelancer profiles: one Green (Mohammad Al-Ghamdi, 79), one Yellow (Noura Al-'Omari, 61), one Building (Fahad Al-Qahtani, 41). SQLite for data + append-only audit log table. Composite score -> tier -> loan recommendation logic.

Module 2 — Claude Explanation Layer (Backend, ~3 hours)

Pre-cache 3 Arabic explanations using PDPL-safe proxy pattern (factor scores only, zero PII). Serve from local JSON on demo day — no live API call. Label output as AI-generated.

Module 3 — Freelancer Mobile Flow (Frontend, ~15 hours)

Arabic RTL, mobile viewport. Screen 1: AI disclosure modal with confirm button. Screen 2: Score card — gauge + tier badge + factor bars + loan recommendation. Screen 3 (decline): Decline reasons + labeled Claude explanation + human review button at TOP in #0052A3. Screen 4 (Building tier): Score improvement roadmap with progress tracker. Consent checkbox with full-width label. Numbers in `<span dir=ltr>`.

Module 4 — Banker Dashboard (Frontend, ~10 hours)

Desktop, dir=rtl, columns in correct Arabic order. Applicant pipeline table. Individual applicant panel: score breakdown + Claude explanation + SIMAH mock status + Approve/Decline/Escalate buttons. Audit log viewer (read-only table).

Module 5 — Digital Proof of Income PDF (~4 hours)

QR-verifiable PDF with Alinma letterhead mock. Income data only — no score, no SIMAH data. One-click generate from banker dashboard.

The 3 Synthetic Profiles to Build

Name	Profession	Score	Tier	Story for judges
Mohammad Al-Ghamdi	App Developer (App developer)	79	GREEN	3 regular clients, consistent income, good savings rate
Noura Al-'Omari	Graphic Designer (Graphic designer)	61	YELLOW	2 clients, variable income, thin savings buffer
Fahad Al-Qahtani	Photographer (Photographer)	41	BUILDING	1 client (71% concentration), low savings — decline with roadmap

8. HOW TO WIN

Judging Criteria Mapping

Criterion	What Judges Want	Mihan Angle
Innovation	New solution to real gap	Only product targeting 1.6M freelancers excluded from ALL bank lending. Not incremental.
Technical Implementation	Working demo, real logic	Live scoring engine, real weight calculations, real Claude Arabic explanations — not slides
Data Analysis	Evidence-based methodology	FICO-inspired factor model with Saudi-specific calibration; Worst-Month x80% stricter than averaging
UX	Polished, usable product	Arabic-first RTL mobile design for freelancers; desktop banker dashboard; WCAG compliant
Real-World Applicability	Would Alinma use this?	Phase 1 under Alinma's existing banking license — no new regulatory approval needed for pilot

4-Minute Pitch Structure

Minute 1 — The Gap

"1.6 million Saudi freelancers hold a Freelance Work Document issued by the Saudi government. Not one commercial bank will lend to them. Not because they're risky — because there's no way to assess them. Mihan fixes that."

Minute 2 — The Product

Walk through Mohammad Al-Ghamdi's application. Show the score card. Show the loan recommendation. Show the Claude-generated Arabic explanation. Show the human review button.

Minute 3 — Why It's Safe

Show the Building-tier decline (Fahad Al-Qahtani) and the improvement roadmap. 'We decline responsibly. We don't overlend.' Show the banker dashboard. Show the audit log. 'Every SAMA requirement — explainability, human review, audit trail — is built in.'

Minute 4 — The Business Case

TAM is SAR 120M/year in platform fees. Phase 1 under Alinma's existing license. Mihan integrates with Alinma's existing Tawarruq structures — no new Sharia approval needed. 'We're ready for a 500-person pilot starting today.'

The 3 Moments That Win the Room

1. The Arabic explanation screen

When judges see Claude generate a real Arabic paragraph explaining the decline — specific to that freelancer's numbers — it lands. Nothing else in the competition will do this.

2. The Building-tier roadmap

Showing that you declined someone and gave them a path forward is more sophisticated than just approving. It signals you understand responsible lending, not just credit scoring.

3. "Under Alinma's existing license"

When a judge asks 'what do you need to go to pilot?' and the answer is 'a signed vendor agreement and SAMA methodology approval — which only Alinma can submit' — you've made them the key to their own product. That's the close.

Things to Avoid

- [X] Do NOT claim Kafalah eligibility — freelancers don't qualify (requires commercial register + 3 years audited financials)
- [X] Do NOT show a live SIMAH API call — it will fail and kill the demo

[X] Do NOT mention Open Banking in Phase 1 — not needed, adds regulatory complexity judges will probe

[X] Do NOT undersell the Building-tier decline — it's your strongest compliance story, not a weakness

[X] Do NOT use placeholder Arabic — Alinma's judges are native speakers and will notice immediately

Post-Hackathon: Alinma Partnership

If Alinma wins the hackathon alongside you and commits to a pilot, items 17 (cloud), 20 (SAMA), 22 (vendor agreement), and 28 (SAMA audit access) become Alinma's problem as much as yours. They have the compliance team, the SAMA relationship, and the infrastructure.

Your exposure narrows to DPIA + DPO — a few weeks of work with the right Saudi consultancy.

FIRST ACTION POST-HACKATHON: Get the pilot MOU or vendor agreement term sheet drafted. Everything else unlocks from that one document.